

Experiment 2: N-Queen Problem

Aim

To solve the N-Queen Problem using Artificial Intelligence techniques.

Objective

- Understand the problem.
- Apply AI search/backtracking.
- Obtain the required solution.

Algorithm

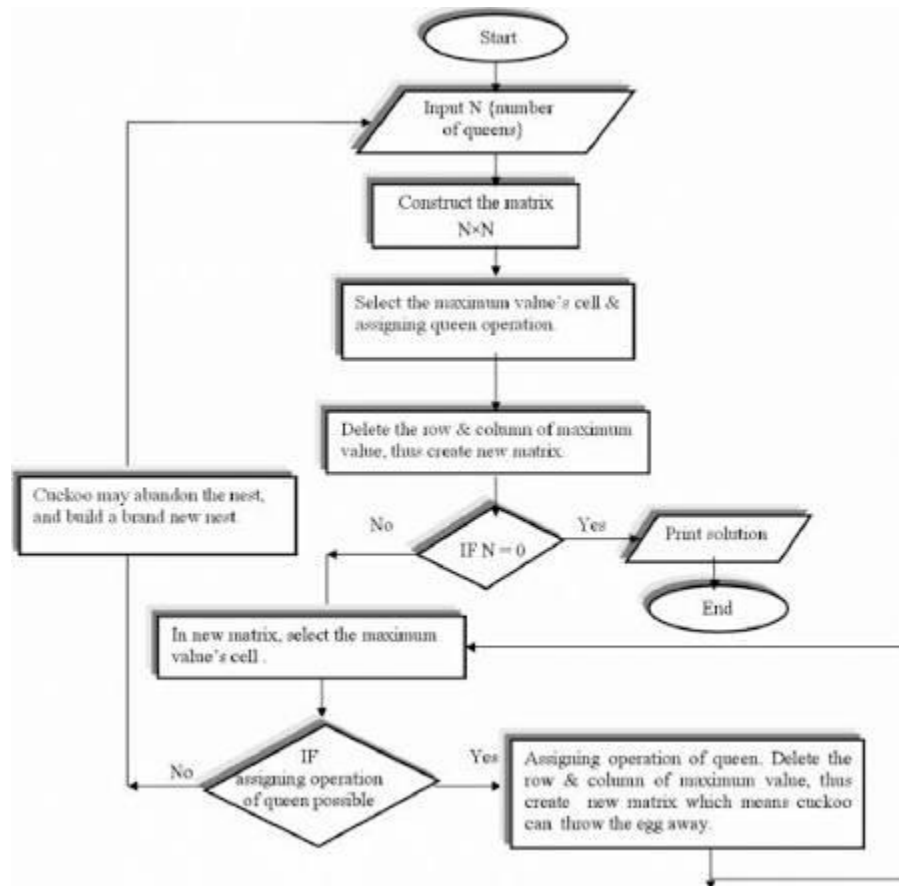
1. Read N.
2. Place queens row by row.
3. Check row/diagonal safety.
4. Backtrack until solution found.
5. Display board.

Program

```
N = int(input("Enter the size of the board (N): "))
board = [[0 for _ in range(N)] for _ in range(N)]
def is_safe(board, row, col):
    # Check left side of the row
    for i in range(col):
        if board[row][i] == 1:
            return False
    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1
    i, j = row, col
    while i < N and j >= 0:
        if board[i][j] == 1:
            return False
        i += 1
        j -= 1
    return True
def solve(board, col):
    if col >= N:
        return True
    for row in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1
            if solve(board, col + 1):
                return True
            board[row][col] = 0
    return False
if solve(board, 0):
    print("Solution Found:\n")
    for row in board:
        print(" ".join("Q" if cell == 1 else "." for cell in row))
else:
    print("No Solution Exists")
```

Program (Python)

Flow chart:



Sample Input

4

Sample Output

One valid 4x4 queen arrangement

Result

N-Queen solved.

Conclusion

The problem was successfully solved using AI techniques.

```
PS C:\Users\athav\AppData\Local\Programs\Microsoft VS Code> & C:\Users\athav\AppData\Local\Programs\Python\Python313\python.exe c:/Users/athav/2.py
```

```
Enter the size of the board (N): 8
```

```
Solution Found:
```

```
Q . . . . . . .  
. . . . . Q .  
. . . . Q . . .  
. . . . . . Q  
. Q . . . . .  
. . . Q . . . .  
. . . . Q . .  
. . Q . . . . .
```

```
PS C:\Users\athav\AppData\Local\Programs\Microsoft VS Code>
```